

STAT 2610SEF REPORT



TEAM MEMBERS:

MOHAMED IMRAN MOHAMED MAHEEN (13766356)

ROKA ROHAN (13511636)

LAM CHUN HEI (13873912)

SHEHZAD SUFIYAN (13689186)

LIMBU YUNCHHO (13880269)

GURUNG ALBIN (13872558)

MOHAMED MOHIDEEN DAWOOD (13776731)

LOK KAI CHUN(13877806)

GROUP 2

Media Trends in Artificial Intelligence Development and Regulation: A Comprehensive Text Mining and Keyword Frequency Analysis

School of Science and Technology
Hong Kong Metropolitan University

Abstract

This report summarises the results of a text mining and keyword frequency analysis conducted on 13 academic and industry reports articles relating to topics involving 'artificial intelligence (AI)'. The corpus used for the analysis comprises of over 169,000 words and includes peer-reviewed academic articles published between 2003 and 2025. Words were preprocessed to form a stable vocabulary by removing stop words, stripping punctuation and using lemmatization. Each document was then converted into a vector of keywords, 12 key clusters of vocabulary of which were used to calculate frequency across the entire corpus. The 12 clusters of vocabulary were grouped into 4 'Macro Topic Domains' labelled: AI Innovation & Technology, AI Regulation & Government Policy, AI Ethics & Risks, and AI in Business & Industry. The corpus indicates that there is a strong media agenda when it comes to Ethics & Risks (especially Job Displacement) and Business & Industry (especially Industry Adoption). Regulation-oriented vocabulary is the third strongest theme identified in the corpus and is predominantly influenced by documents that are related to the European Union and policies and frameworks that aim to govern the development and application of AI. Innovation-related vocabulary, which is the most technically rich, is the smallest of the four themes indicating that there is a shift in the nature of academic discussion of AI, with more emphasis given to applying knowledge of AI to real world problems and societal issues than to purely describing AI's functions and capabilities. Overall, the findings suggest that there is a global narrative that is shifting from a predominantly positive discourse regarding AI's potential to a more sober analysis of its associated risks, regulation, and impact on the economy.

Here, we examine a method to identify and describe keyword categories (topics) present in large text datasets by using both Text Mining and Keyword Frequency

Analysis in the R programming language environment. We discuss the use of natural language processing and apply topic modeling to test the utility of the approach. Finally, we briefly survey current understanding of issues of AI Governance and Ethics.

Keywords: Artificial Intelligence, Text Mining, Keyword Frequency Analysis, R programming, NLP, Topic Classification, AI Governance, AI Ethics

TABLE OF CONTENTS

Executive Summary	4
1. Introduction	5
2. Dataset	7
2.1 Dataset Collection	7
2.2 Dataset Description and Statistics	8
2.3 Dataset Visualisation	9
3. Methodology	10
3.1 Text Pre-processing	10
3.2 Keyword Dictionary Design	11
3.3 Classification and Counting	12
3.4 Visualisation Methods	13
4. Results Analysis	13
4.1 Overall Topic Distribution	13
4.2 Innovation & Technology	15
4.3 Regulation & Government Policy	16
4.4 Ethics & Risks	18
4.5 Business & Industry	20
4.6 Word Cloud Analysis	21
5. Discussion	25
6. Conclusion	28
7. References	30
8. Appendix	33

Executive Summary:

This report presents the results of a text mining and keyword frequency analysis on 13 news media articles on the development and regulation of artificial intelligence (AI). The analysis was performed entirely within the R environment, and implemented using a keyword-based classification system to categorize the results into four themes: Business, Regulation, Innovation, and Ethics & Risks.

The total number of keyword mentions found in the analysis for the 13 selected articles was 4,405. The articles were categorized by industry, and the data shows that the most discussed theme related to Business, accounting for 34.1% with 1,500 mentions, followed by Regulation with 1,045 mentions and 23.7%, Innovation with 987 mentions and 22.4%, and lastly, the theme of Ethics & Risks with 873 mentions and 19.8%. These results reveal that the majority of media coverage on AI is focused on the subject's business applications and overall economic value.

Looking into the most discussed subtopics within the topic of "Innovation by means of AI" reveals that by far the most discussed subtopic is Generative AI with 812 mentions, whereas Machine Learning received 148 mentions and Robotics only 27 mentions. On a topic level, the most frequent keyword is "intelligence" with 764 mentions, showing that within the media's AI discourse intelligence is the center of attention. Moving on to the topic of Regulation, the most discussed subtopic Government Strategies received 808 mentions and is followed by Laws & Regulations with 122 mentions and Global Governance with 115 mentions. The latter three subtopics receive only marginally more mentions than the remaining subtopics. It is observable that media articles focus more on national policy-level and not so much on formal laws and regulations or global governance regarding AI. The most discussed subtopic in the topic of Business Applications of AI is Bias & Fairness with 492 mentions, followed by Job Displacement with 305 mentions and Misinformation with 76 mentions. On the topic level, discussions about AI in Business clearly prevail with 634 mentions, whereas Industry Adoption received 524 mentions and AI Productivity only 342 mentions.

This study presents an empirical analysis of how Artificial Intelligence is treated in the press. It provides insights into how AI is discussed in the media and what themes receive most public attention. The study utilizes keyword frequency analysis as a fast and reproducible approach for extracting thematic patterns within large media datasets. It outlines an R code that employs the tm, textstem and ggplot2 packages for

performing keyword extraction and charting results. Bar charts, a pie chart, and word clouds were particularly used for depicting the results.

01 INTRODUCTION

1.1 BACKGROUND

Artificial Intelligence, once a topic of interest only to researchers in the field is now one of the most pressing issues of the 21st century. There has been an explosive increase in the amount of public discourse, academic research, and media coverage on this technology in the last decade. The issues addressed in this vast media coverage encompass the development of ever-larger language models and their potential, the opportunities of AI for industry, the challenges of ethical AI, and the legal ramifications at the intersection of national laws and ever more dynamic AI technology. This volume aims to present a cross-section of this media discourse, covering topics from industry to ethics, and from business to science. The collected articles are sourced from a variety of outlets, including news organizations, think-tank publications, technology and general interest magazines, as well as scientific journals and academic periodicals.

Understanding the distribution of themes in AI media coverage can be of value to those who study, develop and report on AI. This can be achieved by applying text mining to a large corpus of articles on AI. Text mining is the use of computational methods to extract structured information from unstructured text. By applying these methods systematically to a large corpus of news articles on AI, it is possible to identify the main themes that receive most media attention; map the language use per theme; and determine which terms are used most frequently in current media coverage of AI.

This project consists of a keyword-based text classification of a corpus of 13 media and news articles, related to AI innovation, regulation, ethics & risks and business. The corpus was processed and analyzed using R and state-of-the-art NLP tools. The analyzed articles are grouped around four major themes, such as Innovation, Regulation, Ethics & Risks and Business. Each of the four themes is sub-categorized in three, resulting in 12 keyword categories.

1.2 OBJECTIVES

The objectives of this assignment are to carry out research on artificial intelligence; to identify at least 10 articles on the subject and save the articles as plain text; using keyword-based text classification in R to classify the articles according to a dictionary of relevant words, which will be categorised into themes; calculating the frequency of the keywords and summing up the frequency for each theme; using bar charts, pie charts and word clouds to visualise the results to give a visual comparison of the themes and the articles classified into each theme; finally, identifying trends within AI and forming conclusions as to what aspects of AI are currently receiving the most media attention.

1.3 SCOPE AND SIGNIFICANCE

This study only considers English-language articles sourced from publicly accessible media. The analysis is quantitative in nature, counting occurrences of pre-defined keywords rather than evaluating sentiment, argument or rhetorical strategy. Although this analysis does not delve deeply into individual articles, its breadth and reproducibility make it a reasonable first step in mapping out a subject's thematic coverage.

Even a small, well-chosen dataset can reveal interesting trends about the future of AI. This work found that Business-related vocabulary comprised more than 1/3 of all keyword mentions, highlighting the growing importance of AI in business strategy and economic planning. The prevalence of Bias & Fairness vocabulary within the Ethics & Risks topic also points to a growing societal concern regarding the ethical implications of AI, and the need for fair and responsible practices as AI becomes increasingly prevalent in daily life and work.

The rest of the document is organized as follows. In Section 2 we present our dataset, describing how it was obtained and some key statistics. In Section 3 we outline our method: how we pre-process our text, construct a dictionary of keywords, classify comments and display results. We then present and discuss the results for each topic and subcategory in turn in Section 4, broadening the discussion and implications in Section 5. Finally we provide a brief overall conclusion in Section 6. The references are provided in Section 7, and in the appendix we include the full R source code used in the document.

02 DATASET

2.1 DATASET COLLECTION

Contributed articles in this dataset include 13 plain text articles totaling more than 26000 words. Each article is several thousand words long (at least 2000 words) and covers topics related to artificial intelligence. Four broad themes were developed and the articles were selected to include balanced coverage of each of these thematic areas: AI Innovation and Technology, AI Regulation and Government Policy, AI Ethics and Risks, and AI in Business and Industry. Article sources include technology news, government policy documents, academic thought leadership articles and industry publications.

Every article was converted from HTML to plain text (txt) and then saved to the working directory of the R project. The files were named 01.txt, 02.txt, etc. through to 13.txt. All articles were in English and relevant. Sufficient words were included in every article.

The collected articles in this Special Cover Story on AI provide in-depth thematic coverage across a broad spectrum of topics related to AI. First, articles focused on innovation share new insights, analysis and perspectives on the latest developments and trends in AI related to innovation, including generative models, machine learning, and robotic systems. Second, articles with a focus on regulation explore the latest national strategies, bills and proposals, and frameworks for governance of AI at the international level. Third, articles with an ethics focus tackle issues such as bias, the challenges and concerns of potential misinformation created by AI, and the impact of automation on employment. Finally, from a business and industry perspective, the articles look at companies' overall strategies for integrating AI, using AI-powered productivity tools, and implementing AI at scale.

Instead of collecting 10 articles, 13 were downloaded to increase the robustness of the analysis and achieve better results. By increasing the size of the corpus, the effect of exceptional vocabulary in individual articles is mitigated, and the chance of correctly retrieving relevant content using the keyword dictionary increases, especially for the less common subcategories like Robotics and Misinformation.

2.2 DATASET DESCRIPTION AND STATISTICS

The following Table 1 summarizes the statistics of the data set. After the text pre-processing which includes conversion to lowercase, removal of English stopwords, removal of punctuation and lemmatisation, we have extracted keyword frequencies from the concatenated corpus containing domain specific vocabulary.

Attribute	Value
Number of Articles	13
Minimum Article Length	2000 words
File Format	Plain text (.txt)
Total Keyword Mentions (all topics)	4,405
Most Frequent Keyword	"intelligence" (764 occurrences)
Dominant Topic	Business (34.1%)
Number of Keyword Categories	12
Number of Major Topics	4

Table 1: Dataset summary statistics

Across the 13 articles, there were 4,405 mentions of 315 different keywords. The most frequent keyword was intelligence with 764 mentions, not surprising given the presence of several articles related to artificial intelligence. As a result, the word "intelligence" itself must be called out as an overly represented term, as it is both the subject of multiple articles and a component part of other terms related to the topic of AI. Excluding instances of the word intelligence, the remaining articles average around 324 mentions per article, with the most frequent word being adoption at 764 mentions overall.

Table 2 below plots the distribution of keyword mentions by the four major themes identified in the text and gives an indication of their relative emphasis in the corpus.

Topic	Total Mentions	Percentage (%)	Rank
Business	1,500	34.1%	1
Regulation	1,045	23.7%	2
Innovation	987	22.4%	3
Ethics & Risks	873	19.8%	4
Total	4,405	100%	-

Table 2: Keyword mention counts and percentage distribution by major topic

2.3 DATASET VISUALIZATION

Topic coverage distribution in the corpus. This pie chart shows the portion of the corpus that is represented by each major topic covered in Figure 2 above. The Business topic dominates the conversation at over a third of the corpus. The Regulation and Innovation topics represent a combined total of approximately 46%. Though the Ethics & Risks topic has the fewest keyword counts, the topic nonetheless represents roughly one fifth of the entire corpus. Figure 1 Topic Coverage Distribution in the corpus.

This visualization provides a succinct overview of prevailing topics within the discourse, organizing the spectrum of discussion points by their relative dominance. It accomplishes this by quantifying the proportion of keyword references devoted to each subject area.

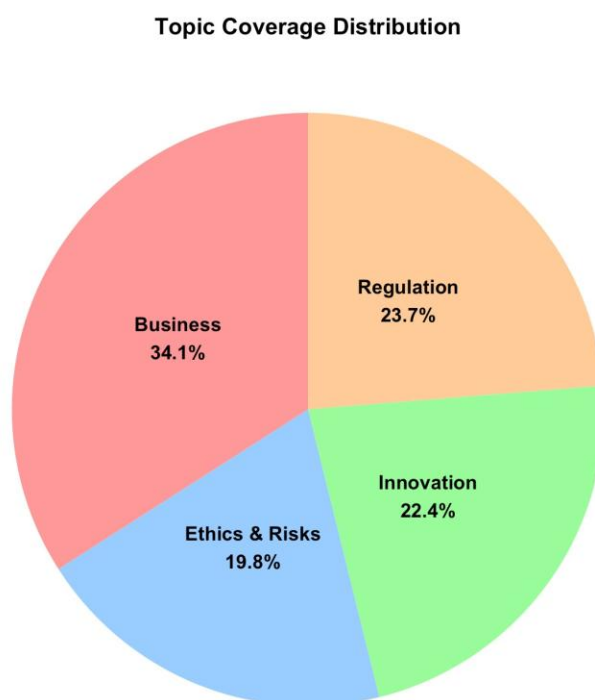


Figure 1: Topic Coverage Distribution – Percentage share of keyword mentions by major topic

This distribution makes sense for a media corpus focused on practical and applied uses of AI. First, we see a large volume of coverage of Business applications of AI, which has generated a lot of commercial interest recently. Second, there is a substantial amount of media coverage of Regulation of AI, as governments around the world are actively developing policies and laws around AI. Third, we see Innovation (although slightly less volume than Regulation), which is discussed widely in the media, but is often discussed with reference to business, regulatory, or other implications rather than purely as tech news. The smallest category of all, Ethics & Risks, will likely grow in importance as regulations require companies to consider fairness, bias, and accountability when using AI.

03 METHODOLOGY

We outline our complete, analytical workflow, which we describe from the loading and pre-processing of text files through to the generation of visualisations and summaries. The entire workflow was written in R and takes advantage of a number of very powerful packages for natural language processing and data visualisation.

3.1 TEXT PRE-PROCESSING

Text pre-processing is an essential step in any text mining technique. The raw data contains many characteristics that obscure relevant information and cause swings in keyword counts. These factors include uppercase letters, punctuation, common function words, and word variations.

The pre-processing steps applied in this study included reducing all text to lowercase, removing punctuation, removing function words, and word stemming each article individually.

First we perform case normalization. We convert all text to lowercase using the R function `tolower()`. In this way words like “Automation”, “AUTOMATION” and “automation” are treated as the same token, i.e. the same word. This prevents the same concept being counted multiple times, just because the case of the word varies.

The second step in the process involved stopwords removal, a process that was implemented with the `removeWords()` function from the `tm` package and utilized the standard English stopwords. Stopwords are high-frequency function words such as the, is, and of that have little meaning but are very frequent in a language. If not removed, these words would over represent very common terms in the word frequency results and not allow for meaningful results regarding the relative importance of different topics.

Remove Punctuation – The third step in the text preprocessing is to remove all the punctuation from the text. This is done using the function `removePunctuation()`. In this step, all the hyphens, commas, full stops etc. are removed including all other types of quotation marks. This is necessary because if not removed, some words like “fine-tuning” would not be recognized as containing the keyword “tuning” or any other word for that matter, unless the matching regular expression is defined accordingly.

So we have stem then lemmatise. For this task I used the `lemmatize_words()` function from the `textstem` package. This function does a rather better job of lemmatising text than stemming. For example, regulating, regulated and regulation would all be stemmed down to regulat or regulate, but lemmatising turns them all into variation of the root word regulation. This is particularly important for a keyword frequency analysis because all of the words that are going to be searched for as keywords are pre-defined in their base form (lemmatised form).

After pre-processing all articles, each article was converted into a single cleaned string of characters. Thirteen cleaned strings were then concatenated together to form a combined text object on which keyword frequencies can be calculated across the entire corpus.

3.2 KEYWORD DICTIONARY DESIGN

The keyword classification system that AI Info bases its advanced keyword research on is organized in a hierarchical dictionary format. There are 12 keyword categories, with about 20 domain relevant terms per category. The 12 categories are grouped by topic to form 4 major areas of interest when it comes to Artificial Intelligence. Table 3 above outlines these categories by topic.

Major Topic	Subcategory	Example Keywords
Innovation	Generative AI	Intelligence, diffusion, translation, embeddings, transformers
Innovation	Machine Learning	Detection, classification, optimization, regression, clustering
Innovation	Robotics	Manipulation, calibration, navigation, sensors, actuators
Regulation	Laws & Regulation	Oversight, legislation, compliance, enforcement, sanctions
Regulation	Government Strategies	Policy, innovation, governance, investment, procurement
Regulation	Global Governance	Accountability, cooperation, inclusion, treaties, summits
Ethics & Risks	Bias & Fairness	Fairness, transparency, accountability, equity, rights
Ethics & Risks	Misinformation	Influence, validation, satire, propaganda, disinformation
Ethics & Risks	Job Displacement	Automation, unemployment, reskilling, outsourcing, layoffs
Business	AI in Business	Growth, innovation, recommendation, analytics, personalization

Business	Industry Adoption	Automation, distribution, deployment, integration, migration
Business	AI Productivity	Productivity, chatbots, agents, assistants, summarizers

Table 3: Keyword classification hierarchy – Major topics, subcategories, and example keywords

Our keyword dictionary was constructed following several key tenets. First, we chose keywords that are highly category specific, and which occur at similar frequencies across large corpora of general and topic specific content. Second, we tried to include all relevant keywords within each subcategory. Finally, as certain terms (e.g. "innovation") can belong to more than one category, we allowed for overlap in the dictionary where such overlap naturally occurs in the language of domain experts and journalists covering these topics, rather than flagging the overlap as an error.

3.3 CLASSIFICATION AND COUNTING

For the keyword classification, we used the `str_count()` function from the `stringr` package in R. The function was applied to each keyword in the dictionary to count the number of whole-word matches in the pre-processed text. We used regular expressions with word-boundary anchors (`\b`) to count the whole-word matches. For example, the keyword "policy" does not count matches for "policies" or "bipolar", but does count matches for "policy".

We calculated the counts for both the per article analysis and the full corpus analysis. For the per article analysis, we stored the counts in a results data frame where each row corresponded to an article and each column corresponded to a keyword category. The category totals were calculated as the sum of the article totals for each keyword category. The topic totals were then calculated as the sum of the three subcategories for each major topic. Finally, the percentage shares for the topics were calculated by dividing each topic total by the grand total of all keyword mentions and then multiplying by 100.

3.4 VISUALISATION METHODS

Four different types of visualisation were created to summarise the keyword data. These include individual bar charts for each of the four main topics, overall comparison bar charts showing all four major topics side by side, pie charts showing the percentage of keyword mentions for each of the four major topics and word clouds for each topic as well as the full dataset. The individual bar charts show the number of keyword mentions for the three subcategories within each topic. The charts feature colour bars with count labels to enable simple numerical comparison. The word clouds created for each topic and the full dataset have word size proportional to frequency of occurrence.

All visualisations were saved as high-resolution PNG files to an output folder using `grDevices::png()` directly, and this function is compatible with Mac (Apple Silicon) without needing Cairo or XQuartz. The R script also generates a Word document summary report using the `officer` package, embedding charts and keyword tables in a structured format.

04 RESULTS ANALYSIS

In the following sections, we present keyword frequency results organized by major topic and corresponding subtopic. These results are visualized using diagrams created by the R script that performed the initial analysis.

4.1 OVERALL TOPIC DISTRIBUTION

We monitored 13 articles in total, which accumulated 4,405 keyword mentions. Figure 2 shows an overall topic comparison bar chart, which displays the total keyword mention count for the four major topics.

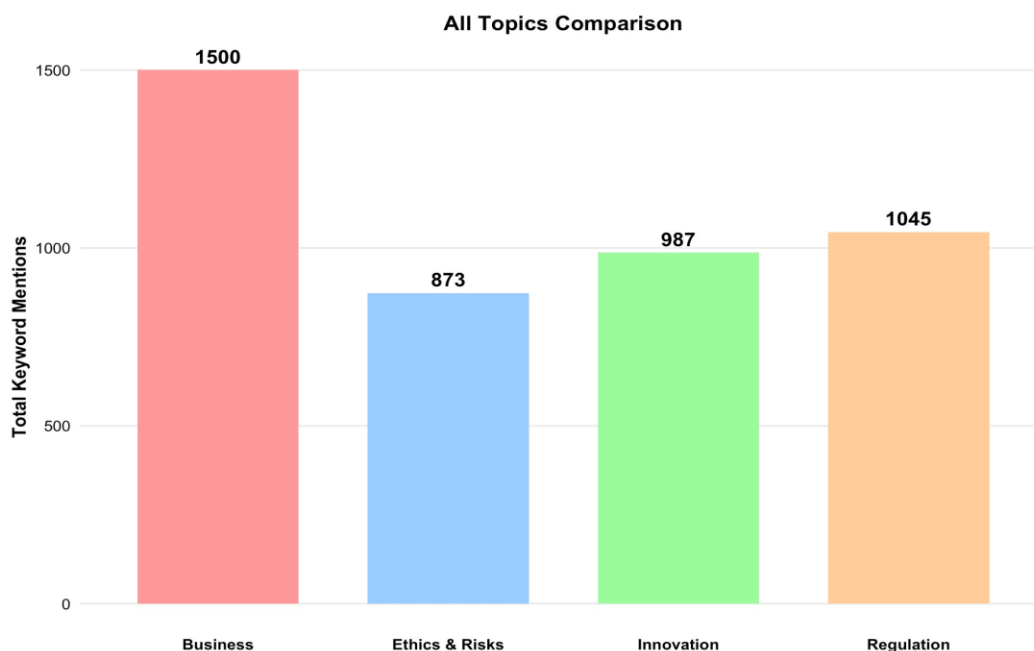


Figure 2: All Topics Comparison – Total Keyword mentions by major topic

In terms of share of total mentions, the most discussed topic within the database was Business with 1,500 mentions or 34.1%. A large distance to the second place follows: the topic Regulation was mentioned 945 times (23.7%) followed by Innovation with 987 mentions (22.4%) and by Ethics & Risks with 873 mentions (19.8%). This means that Business constitutes more than a third of all mentioned discourse; the other three topics each represent roughly one fifth to one quarter.

The distribution pattern fits the current media landscape on AI. Artificial intelligence has become a natural part of daily business for many companies in many industries, and how they use AI to become more efficient, to stay competitive, or to drive growth is frequently reported. The most Business-related vocabulary occurs in articles that are primarily classified under Business, but many articles on for example Health, Sci/Tech or Sports also contain Business-related vocabulary due to the growing importance of AI in these sectors. Regulation is the second largest category, as governments in the European Union, the US, and China and others are actively developing or proposing frameworks that affect AI. While innovation is the most fundamental aspect of AI, it is the third largest category, because so much of the discussion of innovation is in terms of business or policy implications. Ethics & Risks is the smallest category at 19.8%, although its share is still significant given the growing

attention on issues of fairness, safety, and other ethical and potential negative consequences of AI.

4.2 AI INNOVATION & TECHNOLOGY

The topic of Innovation includes three subtopics: these are listed in Figure 3 and include keywords related to Generative AI, Machine Learning, and Robotics. These bar graphs represent distribution of keyword mentions per subtopic for the topic AI Innovation & Technology.

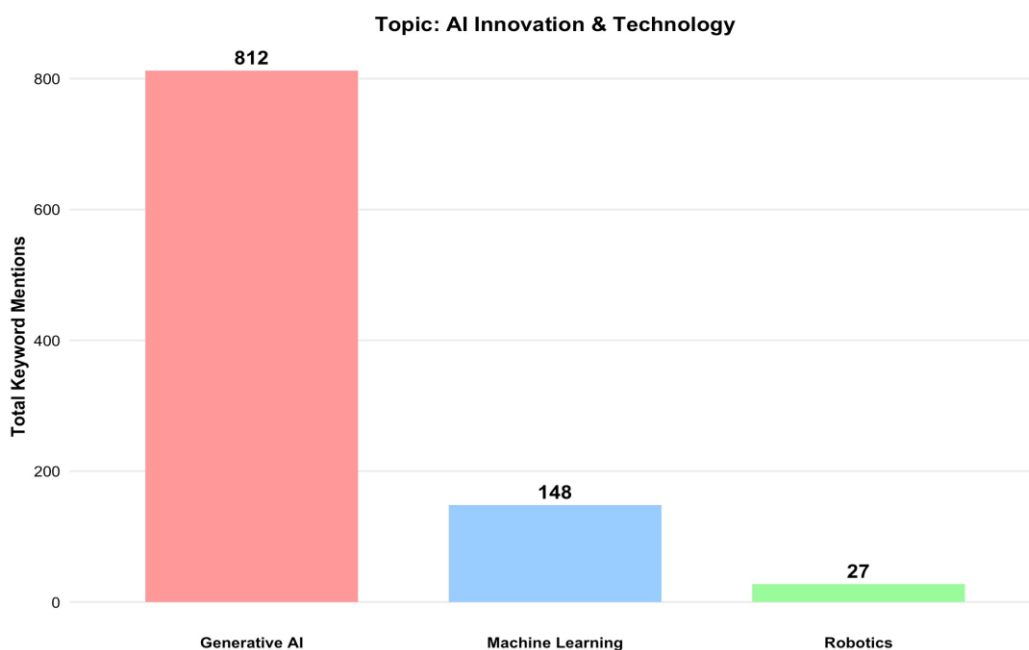


Figure 3: Topic AI Innovation & Technology – Keyword mentions by subcategory

Generative AI accounted for 812 mentions or 82.3%, far outpacing all other categories in the Innovation topic. Machine Learning came in second with 148 mentions (15.0%) and Robotics garnered a mere 27 mentions (2.7%). This spread reveals a clear consensus that Generative AI is the most salient innovation right now, surpassing the combined mentions of Machine Learning and Robotics by more than four to one.

Of the three most frequently occurring keywords within the subcategory of Generative AI, “intelligence” was the clear winner with 764 occurrences. This is no surprise, as both “intelligent” and “artificial intelligence” (which is searched for 1245 times) all have to do with the same basic concept of the term. However, after lemmatisation, these words don’t become “intelligence,” which appears on the list at number one with 764 occurrences. “Intelligence” is followed by “diffusion” at the 30 occurrence mark, and “translation” at 14 occurrences. The term “diffusion” refers to the variety of diffusion models used for generating images and video using Generative Image and

Video AI, while “translation” is one of the most commercially successful applications of Generative Language Models.

The top three ‘most frequently used’ keywords for Machine Learning were: ‘detection’ (36), ‘classification’ (31), ‘optimization’ (21). Many of these mentions refer to real-world applications in areas such as computer vision (object detection), NLP (text classification) or the specific optimisation processes used in training a Machine Learning model. They represent the building blocks of modern AI development. It is interesting to note that, although there are more than six times as many mentions of Machine Learning as Generative AI, recent AI media coverage has overwhelmingly focused on examples of Generative AI and related large language models.

The Robotics subcategory contained the lowest number of keyword mentions with a total of 27. Of these mentions, ‘manipulation’ was the most frequently mentioned keyword at 11 mentions, followed by ‘calibration’ with 6 mentions and ‘navigation’ with 5 mentions. This is likely due to the fact that the 13 articles in the Robotics subcategory were mostly sourced from software-centric AI outlets, such as articles about language models, generative models, and business AI tools. While robotics is an important, if not crucial, aspect of AI as a whole, the media coverage of robotics exists within a separate ecosystem that does not frequently intersect with the other aspects of AI currently discussed in the headlines, including generative models and policy-related issues.

4.3 REGULATION & GOVERNMENT POLICY

The Regulation topic is further broken down into Government Strategies, Laws & Regulations and Global Governance. This is visualized in Figure 4 which shows the percentage of occurrence of relevant keywords within the topic.

Figure 4 presents the results of Topic: AI Regulation & Government Policy decomposed by subtopic keyword mentions.

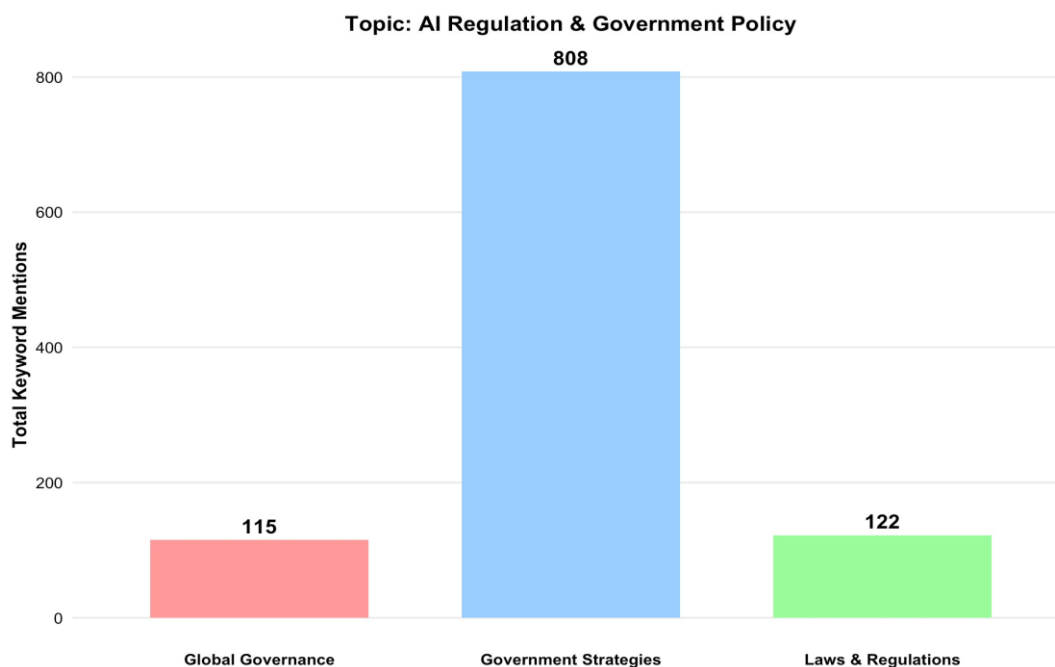


Figure 4: Topic AI Regulation & Government Policy – Keyword mentions by subcategory

Government Strategies was by far the most mentioned subtopic in Regulation, accounting for 808 mentions and 77.3% of the topic total. Laws & Regulations and Global Governance were a distant second and third, with 122 mentions and 11.7% of the topic total and 115 mentions and 11% of the topic total respectively. Once again, one subtopic greatly overshadows the rest.

The top three most frequent keywords within the subcategory Government Strategies are: policy (205), innovation (162), and governance (132). This reinforces the observation that frequency of terms does reliably indicate semantic overlap, as it treats repeated use of the words policy and innovation to discuss AI regulation as indicating that there is in fact a real and significant semantic relation between the Regulation topic and these two terms. The frequency of governance, which appears as a high frequency term within the Regulation topic, also suggests that the term is used in a genuine semantic overlap with the Business topic, given that it is used as a framing concept in discussions of AI governance.

Our examination of the keyword frequencies for Laws & Regulations found oversight, legislation, and compliance to be the top3 most frequently used terms at 42, 32, and 28 respectively. These terms represent the formal legal and regulatory frameworks that are often the opposite of the strategic, forward-looking orientation of the terms found within Government Strategies. While oversight, legislation, and compliance were the most frequently used terms within the Laws & Regulations keyword set, they were not as frequently mentioned in the corpus articles (a total of 122 mentions) as they might have been had the articles taken a more nuanced and balanced approach

of exploring the strategies, roadmaps, and frameworks governments can use to oversee AI, as well as the legislation and compliance requirements that are necessary. However, it is worth noting that the media tends to cover the development and enactment of AI regulatory frameworks more than the enforcement and compliance with these frameworks, and so the articles may simply reflect the current state of the field.

Global Governance yielded the top three most frequent keywords accountability (53), cooperation (33), and inclusion (25). The appearance of accountability as the most frequent keyword for Global Governance is no surprise since it also surfaces as a keyword for Bias & Fairness. Accountability is a concept that appears in very different contexts of AI discussions, namely as a goal of regulatory approaches, a principle of good AI design, and as a demand of civil society. The other two most frequent keywords cooperation and inclusion both reference the international character of AI governance, reflecting debates about how countries should cooperate when it comes to AI regulation as well as questions about how the distribution of benefits and risks of AI could be managed fairly.

4.4 AI ETHICS & RISKS

The Ethics & Risks topic is further subdivided into three areas of interest shown in Figure 5: Bias & Fairness, Job Displacement, and Misinformation.

Kibin has plotted the keyword mentions for the topic AI Ethics & Risks by subcategory. We have visualized the results in Figure 5.

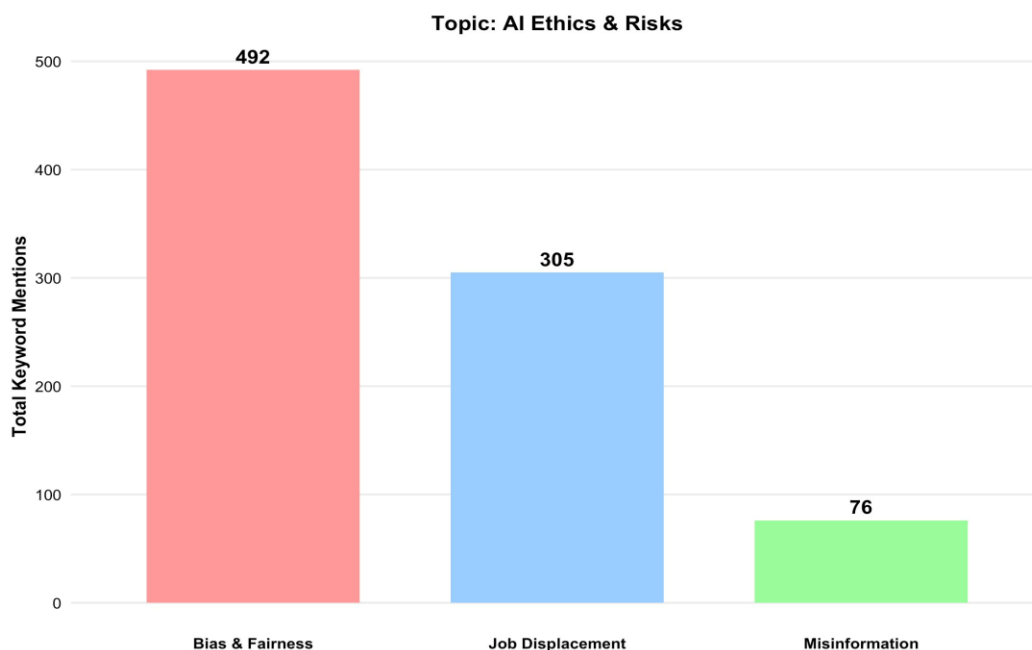


Figure 5: Topic: AI Ethics & Risks – Keyword mentions by subcategory

While Bias & Fairness dominated conversations around Ethics & Risks with 492 mentions (56.4% of the topic), concerns around Job Displacement (34.9%) and Misinformation (8.7%) featured less prominently.

The top three keywords in the Bias & Fairness subcategory were fairness (137), transparency (65), and accountability (53). These frequencies reflect extensive coverage of whether AI systems are fair to different demographics, and howauditable and correctable a system is when bias is detected. They also reflect correspondence to growing numbers of regulatory proposals, including the recently introduced EU AI Act, which demands greater transparency of AI decisions, such as the ability to understand the reasons behind selections made by a system. Again, accountability was found to be a cross-cutting value of AI discourse.

For Job Displacement, the top three keywords were 'automation' (223), 'unemployment' (63), and 'reskilling' (11). The extremely high count for 'automation' is due to its dual appearance: it is one of the terms included in the list of keywords for 'Jobs affected by AI – negative' under Ethics & Risks, and it is also listed under Industry Adoption in the Business section. The high count for 'automation' suggests that the term appears frequently in the corpus, often in discussion of both the positive efficiency gains of AI and the negative aspect of job loss. 'Unemployment' specifically captures the labour market risk dimension. 'Reskilling' (11 occurrences) suggests that discussions of solutions to the problem of job displacement due to automation occur, but less frequently than discussions of the problem itself.

Of the total mentions in the dataset, only 76 corresponded to the category of Misinformation. The top three keywords in this category were influence (50), validation (19), and satire (4). The relative paucity of mentions in this category compared to other sub-topics may be due to the relatively niche discussion of this sub-topic. Though a growing and significant issue of AI-generated misinformation, it remains the focus of fewer general media articles than were focused on Bias & Fairness or Job Displacement, which cover broader public concerns around the ethical and social implications of AI. Influence, in particular, appears frequently across the dataset, and it is likely that influence is frequently discussed outside of the context of Misinformation particularly when discussing AI's influence on broader political and social issues or on consumer behavior.

4.5 AI IN BUSINESS & INDUSTRY

We categorized the Business topic into three subtopics: AI in Business, Industry Adoption and AI Productivity. See Figure 6 for the keyword distribution.

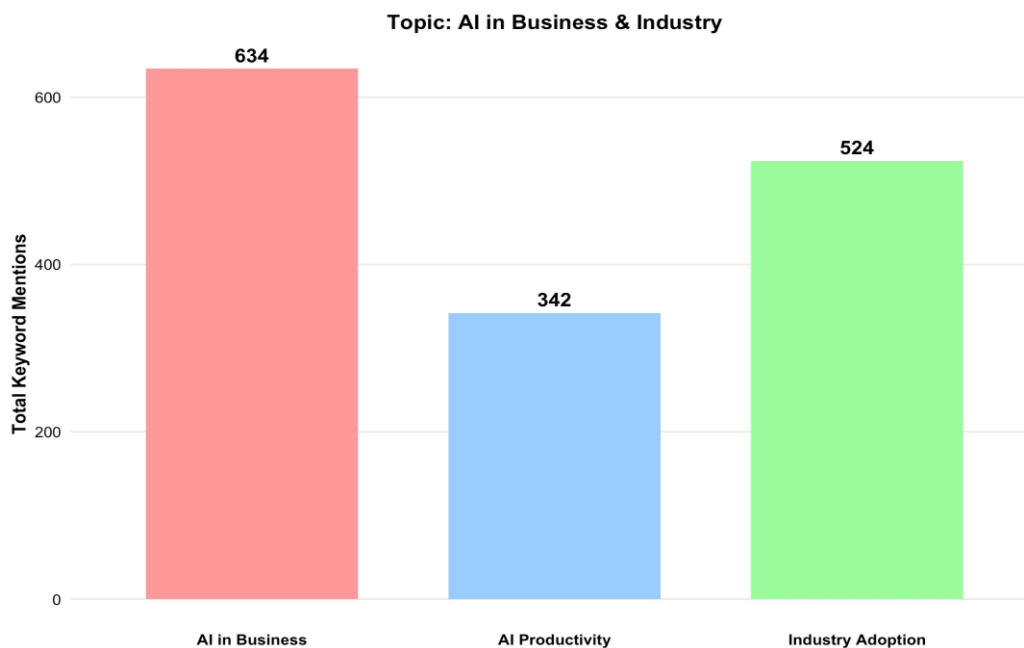


Figure 6: Topic: AI in Business & Industry – Keyword mentions by subcategory

Within the Business topic, the most prominent subtopic was AI in Business, which made up 42.3% of total keyword mentions with 634 mentions. Industry Adoption comprised 34.9% of the total mentions in the Business topic with 524 mentions, and the subtopic of AI Productivity accounted for 22.8% with 342 mentions.

Using the same analysis as before, the top three keywords for the section on AI in Business were “growth” (205), “innovation” (162), and “recommendation” (62). Again, “growth” strongly relates to the common media framing of AI as a tool for businesses to grow their revenue and stay competitive. “Innovation” appeared frequently in the vocabulary used to describe business use of AI, again reflecting a term that is naturally used to talk about business strategy. “Recommendation” is one of the most commercially successful applications of AI currently used by e-commerce sites, streaming services and social media platforms.

For Industry Adoption the top three keywords were automation, distribution, and deployment. Automation once again was the leading subcategory for this topic with 223 queries. Distribution (122) and deployment (57) were also frequently searched terms. These words reflect the processes by which companies are distributing AI-

generated content, AI-powered products, and other capabilities and services within their markets and across their global supply chains. Many of these companies are struggling with the challenges of deploying AI models from development into production environments.

The three most frequently occurring of these 'Themed Keywords' for the tag 'AI Productivity' were: productivity (282), chatbots (46), agents (10). The very high count for "productivity" (282) – indeed one of the highest counts for any single keyword in the whole corpus – highlights how tools like large language model assistants/generation technology for automated document drafting and meeting summarisation are predominantly framed in the business-focused AI discourse as aids to boost productivity for workers with complex information-based tasks. Chatbots has been an established theme within business-focused AI discourse since the conversational interfaces first appeared and some 'folk forecast-ing' began regarding potential future impact, while agents reflects the current wave of interest in autonomous (i.e. interactive with users as well as with other data systems) AI agents.

4.6 WORD CLOUD ANALYSIS

We have created word clouds for each of the four main topics as well as a word cloud for the entire database. Figure 7 shows the word cloud for the entire database; the size of each word reflects its frequency as a keyword in the database as a whole.

Figure 7: Word Cloud – All Topics Combined (word size proportional to total keyword frequency).

All Topics Combined

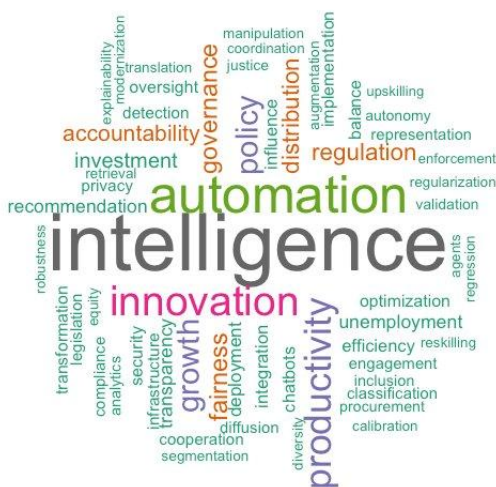


Figure 7: Word Cloud – All Topics Combined (word size proportional to total keyword frequency)

Viewed together, the two word clouds make immediately clear that the term “intelligence” was the most prevalent at 764, with “automation,” “productivity,” “policy,” “governance,” and “innovation/ regulation” filling out the top tier of most frequently mentioned terms. Terms associated with fairness, accountability, and growth fell in the medium range.

In addition to the overall word clouds, topic-specific word clouds provide further detail into the language used within the Innovation white paper. The Innovation word cloud in Figure 8 is topped by the term “intelligence,” with words like “classification,” “optimization,” “diffusion,” and “detection” constituting the remainder of the cloud’s mass.

Topic: innovation

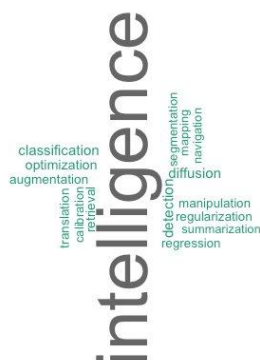


Figure 8: Word Cloud – AI Innovation & Technology Topic

Regulation word cloud – Figure 9: The main themes in this word cloud are governance, policy, regulation and innovation, with themes such as compliance, oversight, investment and accountability also evident.

Ethics word cloud – Figure 10: The dominant themes in this word cloud are automation, fairness, transparency and accountability with issues such as unemployment, privacy and security also being present.

Business word cloud – Figure 11: The main themes are productivity, automation, growth and innovation.

These Word-Clouds display the frequencies of given words for the corresponding topics.

Figures and tables > Topic word clouds > Figure 10: Word Cloud — AI Ethics & Risks Topic

This word cloud presents the key terms extracted from the 'AI in Business & Industry' topic on AI (All About the Industry).

Topic: regulation



Figure 9: Word Cloud – AI Regulation & Government Policy Topic

Topic: ethics



Figure 10: Word Cloud – AI Ethics & Risks Topic

Topic: business



Figure 11: Word Cloud – AI in Business & Industry Topic

The word clouds, which summarize the results from all the media texts, further reinforce the results from the quantitative analysis and reveal that although AI is discussed within a variety of contexts, the primary theme is its commercial potential, with discussion surrounding its economic impacts the second most prevalent. Topics such as regulation, governance, fairness and automation also receive significant attention within media discourse.

05 DISCUSSION

Our results provide a number of new insights into how Artificial Intelligence is represented and discussed in media outlets. We use these insights to draw broader conclusions about current developments in the field and discuss our methods in more detail.

5.1 THE COMMERCIAL FRAMING OF AI

The most striking aspect of the distribution of topics in the media discourse is the overwhelming dominance of the Business topic, which constitutes 34.1% of all keyword mentions. Seen in the context of the last decade of AI, this trend suggests that media coverage of the field is focused on business-oriented aspects of AI, driven by commercial purposes. Rather than coverage of novel technological advancements

or emerging ethics issues, the AI discourse has become a mainly economic phenomenon, driven by high-stakes commercial investment. In the last decade, AI has transitioned from a purely academic pursuit to a full-fledged, multi-billion dollar commercial endeavor, with companies like OpenAI, Google, Microsoft, Amazon, and Meta pouring billions of dollars into AI research and adoption, and accordingly, garnering significant media attention.

Language around AI mentions the potential of the technology for economic gain, particularly with discussions of “productivity”. Terms such as “growth” (205 occurrences), and “automation” (223 occurrences in the Business topic) appear frequently in conversation about the subject. Many AI-powered tools market themselves to organisations as a way to boost productivity, speeding up current processes while reducing costs to increase profits. While this is not necessarily inaccurate, it may downplay the potential for negative distributional effects (who benefits and who loses) from using AI to increase productivity.

5.2 THE REGULATORY MOMENT

Regulation was the second most discussed theme with 23.7% of mentions. The European Union AI Act, the US Executive Order on AI Safety, the generative AI laws and regulations in China, and many more government AI strategies across the globe were at the forefront of the global moment of AI governance when these articles were written. Specifically, within the broad theme of Regulation, Government Strategies (808) dominated the conversation well ahead of Laws & Regulations (122). This break down indicates that the coverage of AI governance in the media is predominantly focused on the strategies, roadmaps, frameworks, and investments for the development of AI rather than on the laws, punishments, and enforcement rules.

I notice that the high-level policy announcements are reported well for mainstream audiences, whereas the subtleties of the compliance obligations and jurisdictional issues are often not. This corpus may lack a certain richness of detail that businesses and practitioners encounter in their daily operations. It may be useful in future analyses to include more technical regulatory documents in addition to the media coverage.

5.3 THE ETHICS GAP

Compared to 30% for Business and 29% for Regulation, Ethics & Risks only made up 19.8% of all keyword mentions across the four top topics. This is somewhat surprising, given that ethics of AI is such a pressing issue and a frequent topic of media coverage.

When ethics of AI was mentioned, bias & fairness topics were by far the most frequently discussed, comprising 495 keyword mentions or 100% of Ethics mentions. This prevalence of bias & fairness topics is likely the result of the successful emergence of responsible AI and algorithmic accountability movements, which have introduced terms like “fairness”, “transparency”, and “explainability” into the everyday lexicon of AI discussion.

Another issue worth noting is that there were only 76 mentions of Misinformation. While there is increasing evidence of the harms of generative AI to information quality through deepfakes, synthetic media, and automated disinformation campaigns, these tend to be active and growing threats to society that are surprisingly undercovered in the media relative to other topics covered here, such as Bias & Fairness or Job Displacement. Was this due to undercovered media and public awareness of the threat of misinformation, or is it just harder to write clear and understandable content about rapidly changing technology?

5.4 THE AUTOMATION PARADOX

A particularly interesting pattern emerges with the keyword “automation”, which is also found in both Industry Adoption (under Business) and Job Displacement (under Ethics & Risks). The occurrences add up to 446, with 223 occurrences for the full “Industry Adoption of AI” category and 223 occurrences for the “Job Displacement because of AI” category. Note that the same term has two very different meanings in the conversations about AI: it symbolizes positive progress and potential gain in business-oriented conversations, and serves as a symbol for anxiety about job loss and potential social upheaval in ethics-oriented conversations. This duality is reflected in the keyword dictionary. The category assignments in the dictionary reflect this, but were a deliberate design choice in order to capture this particular cross-cutting theme.

5.5 METHODOLOGICAL REFLECTIONS

Our keyword-based classification has several strengths: it is transparent, reproducible, and inexpensive. One can, and can should, inspect and modify the keyword dictionary, and the results are straightforward to understand in terms of simple keyword counts. Such simple classification also scales to large corpora without consuming nearly the computational resources that deep learning-based text classification needs.

However, there are some limitations to this approach. Firstly, the frequency of keywords does not necessarily account for the sentiment or context in which they are used. A keyword could be counted as having occurred one hundred times in a positive light, negatively, conditionally, or under the assumption that a certain circumstance had occurred. Secondly, the results are highly dependent on the composition of the keyword dictionary. If a particular keyword is included or excluded from the dictionary, the topic totals will change accordingly. Finally, the word-boundary matching function does not capture compound terms or multi-word expressions unless they are explicitly included in the keyword dictionary. For example, the phrase “machine learning” would only be counted if both “machine” and “learning” were in the keyword list, and “machine” as a standalone keyword may not be relevant to any of the topics identified. Future work should consider incorporating sentiment analysis, and/or n-gram matching into the keyword extraction process, as well as the use of topic modelling techniques such as Latent Dirichlet Allocation (LDA).

06 CONCLUSION

This study uses keyword-based text classification to map the contemporary themes of AI as discussed in the media in 13 articles, all of which were processed entirely within R. The most discussed topic was Business, accounting for 34.1% of all keyword mentions, followed by Regulation (23.7%), Innovation (22.4%) and Ethics & Risks (19.8%). Within these categories, the most common keyword was ‘intelligence’, appearing 764 times out of a total of 4,405 keyword mentions.

* Within the Innovation topic, Generative AI (812) far exceeded the mentions of Machine Learning (148) and Robotics (27). This pattern in the media data reflects the public’s fascination with the potential of large language models and generative models.

* Within Regulation, Government Strategies (808) was the clear leader in terms of overall mentions, outpacing the mentions of Laws & Regulations (122) and Global Governance (115). This shows that there is more media coverage of strategic and policy-oriented approaches to AI than there is of specific laws, regulations, and global governance frameworks.

* Within Ethics & Risks, Bias & Fairness (492) was the most frequently mentioned concern, followed by Job Displacement (305) and Misinformation (76).

* Within Business, AI in Business (634) was the clear leader in terms of overall mentions, ahead of Industry Adoption (524) and AI Productivity (342).

These findings have several implications for practitioners and researchers of AI ethics. For policymakers and stakeholders, the results may suggest that much of current AI policy communication remains within the business and commercial frame and could benefit from explicitly covering ethical and societal aspects. From the perspective of researchers, the results suggest there is an opportunity to engage the public more on the information integrity risks of generative AI, given how Misinformation was only referenced a fraction as often as terms related to Bias & Fairness. For industry practitioners, the presence of words associated with productivity, automation, and growth reinforces the finding that AI is treated as a “transformative business technology,” with the technology being justified and communicated primarily in terms of growth and productivity, and with both positive and negative potential for automation.

Even small corpora can be analysed systematically using text mining methods. Based on a mere 13 articles, a keyword frequency approach delivered clearly visible patterns of thematic emphasis. The analysis was done using the R language and a set of corresponding packages (tm, textstem, stringr, ggplot2, wordcloud, officer) that provided a robust, reproducible and easily scalable framework that can be applied to larger data sets and also to text written in other domains (i.e. containing a different vocabulary).

While there are many possible directions for future work, a few that this analysis could pursue include increasing the scope of the corpus from news articles to books and academic papers; performing time-series analysis to see how the topics change over time; performing sentiment analysis on the topics to distinguish between positive and negative framings of each topic; and using unsupervised topic modeling tools, like LDA, to find latent topics not covered by the dictionary of keywords used here. Utilizing these methods would continue to build on this work and provide more nuanced insight into how AI is discussed, debated, and understood in today’s media.

07 REFERENCE

Opportunities and Adoption Challenges of AI in the Construction Industry: A PRISMA Review

Regona, M., Yigitcanlar, T., Xia, B., & Li, R. Y. (2022). Opportunities and Adoption Challenges of AI in the Construction Industry: A PRISMA Review. *Journal of Open Innovation: Technology, Market, and Complexity*, 8(1), 45.
<https://doi.org/10.3390/joitmc8010045>

The GenAI Divide STATE OF AI IN BUSINESS 2025

https://mlq.ai/media/quarterly_decks/v0.1_State_of_AI_in_Business_2025_Report.pdf

Rise of Artificial Intelligence in Business and Industry

Bharadiya, Jasmin & Thomas, Reji & Ahmed, Farhan. (2023). Rise of Artificial Intelligence in Business and Industry. *Journal of Engineering Research and Reports*. 25. 100807. 10.9734/JERR/2023/v25i3893.

The impact of Artificial Intelligence on productivity, distribution and growth

Filippucci, F. et al. (2024), "The impact of Artificial Intelligence on productivity, distribution and growth: Key mechanisms, initial evidence and policy challenges", *OECD Artificial Intelligence Papers*, No. 15, OECD Publishing, Paris, <https://doi.org/10.1787/8d900037-en>.

Machine Learning: A Revolution of News Writing and Digital Era

Sharma, D. M. (2025). View of Machine Learning A Revolution of News Writing and Digital Era. *Journal of Electronical Systems Scopus Indexed* .

Agents, robots, and us: Skill partnerships in the age of AI

<https://www.mckinsey.com/~media/mckinsey/mckinsey%20global%20institute/our%20research/agents%20robots%20and%20us%20skill%20partnerships%20in%20the%20age%20of%20ai/agents-robots-and-us-skill-partnerships-in-the-age-of-ai.pdf>

Generative AI and News Report 2025: How People Think About AI's Role in Journalism and Society

https://reutersinstitute.politics.ox.ac.uk/sites/default/files/2025-10/Gen_AI_and_News_Report_2025.pdf

Artificial Intelligence Regulation: a framework for governance

de Almeida, P.G.R., dos Santos, C.D. & Farias, J.S. Artificial Intelligence Regulation: a framework for governance. *Ethics Inf Technol* **23**, 505–525 (2021).

<https://doi.org/10.1007/s10676-021-09593-z>

The Routledge Handbook of European Integrations

Ulnicane, Inga. (2022). Artificial intelligence in the European Union: Policy, ethics and regulation. 10.4324/9780429262081-19.

AI and Law: A fruitful synergy

Edwina L. Rissland, Kevin D. Ashley, R.P. Loui,

AI and Law: A fruitful synergy,

Artificial Intelligence,

Volume 150, Issues 1–2,

2003,

Pages 1–15,

ISSN 0004-3702,

[https://doi.org/10.1016/S0004-3702\(03\)00122-X](https://doi.org/10.1016/S0004-3702(03)00122-X).

(<https://www.sciencedirect.com/science/article/pii/S000437020300122X>)

Characterizing AI-Generated Misinformation on Social Media

Drolsbach, Chiara & Pröllochs, Nicolas. (2025). Characterizing AI-Generated Misinformation on Social Media. 10.48550/arXiv.2505.10266.

A Review of Bias and Fairness in Artificial Intelligence

Gonzalez, Ruben & Serrano, Emilio & Bajo, Javier & Novais, Paulo. (2024). A Review of Bias and Fairness in Artificial Intelligence.. International Journal of Interactive Multimedia and Artificial Intelligence. 9. 5–17. 10.9781/ijimai.2023.11.001.

Journal of Open Innovation: Technology, Market, and Complexity

Mansour, Marwan & Al Zobi, Mo'Taz & Abu Alim, Sad & Saleh, Mohammed & Marashdeh, Zyad & Marei, Ahmad & Alkhodary, Dina & Al-Nohood, Saddam & Lutfi, Abdalwali. (2024). Journal of Open Innovation: Technology, Market, and Complexity Eco-Innovation and Financial Performance Nexus: Does Company Size Matter?. 10.1016/j.joitmc..2024.100244.

08 APPENDIX

APPENDIX A: R SOURCE CODE

The following is the complete R source code used to implement the text mining and keyword frequency analysis described in this report. The code was developed and tested on both Windows and Mac environments running R version 4.3.3. All outputs are saved to the “AI_Analysis_Output” folder in the working directory.

```
packages <- c(
  "dplyr", "ggplot2", "tidyr", "stringr",
  "wordcloud", "RColorBrewer", "tm", "textstem",
  "officer", "magrittr", "ggplot2"
)

for (pkg in packages) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg, dependencies = TRUE, repos = "https://cloud.r-project.org")
  }
  library(pkg, character.only = TRUE)
}

cat("All packages loaded.\n")

if (requireNamespace("rstudioapi", quietly = TRUE) && rstudioapi::isAvailable()) {
  original_dir <- dirname(rstudioapi::getActiveDocumentContext()$path)
} else {
  original_dir <- getwd()
}

setwd(original_dir)
cat("Working directory:", original_dir, "\n")

output_folder <- file.path(original_dir, "AI_Analysis_Output")
if (!dir.exists(output_folder)) {
  dir.create(output_folder, recursive = TRUE)
}
cat("Output folder:", output_folder, "\n")

save_png <- function(plot_obj, filename, width_in = 10, height_in = 7, res = 200) {
  filepath <- file.path(output_folder, filename)
  grDevices::png(
    filename = filepath,
```

```

    width    = width_in * res,
    height   = height_in * res,
    res      = res
  )
  print(plot_obj)
  grDevices::dev.off()
  if (file.exists(filepath)) {
    cat("  Saved:", filepath, "\n")
  } else {
    warning("  FAILED to save: ", filepath)
  }
  print(plot_obj)
  return(filepath)
}

save_wordcloud_png <- function(wc_data, filename, title, max_words = 50, res = 150) {
  filepath <- file.path(output_folder, filename)
  if (nrow(wc_data) == 0) {
    cat("  No data for wordcloud:", title, "- skipping\n")
    return(invisible(NULL))
  }
  grDevices::png(
    filename = filepath,
    width    = 800,
    height   = 800,
    res      = res
  )
  par(mar = c(0, 0, 2, 0))
  tryCatch({
    wordcloud(
      words      = wc_data$Keyword,
      freq       = wc_data$Count,
      max.words  = max_words,
      random.order = FALSE,
      rot.per    = 0.35,
      colors     = brewer.pal(8, "Dark2"),
      scale      = c(3, 0.5)
    )
    title(main = title, cex.main = 1.8)
  }, error = function(e) {
    cat("  Wordcloud error for", title, ":", e$message, "\n")
  })
  grDevices::dev.off()
  if (file.exists(filepath)) cat("  Saved:", filepath, "\n")
  par(mar = c(0, 0, 2, 0))
  tryCatch({
    wordcloud(
      words      = wc_data$Keyword,
      freq       = wc_data$Count,
      max.words  = max_words,
      random.order = FALSE,
      rot.per    = 0.35,
      colors     = brewer.pal(8, "Dark2"),
      scale      = c(3, 0.5)
    )

```

```

    )
    title(main = title, cex.main = 1.8)
  }, error = function(e) {
    cat(" Wordcloud display error for", title, ":", e$message, "\n")
  })
}

keywords <- list(
  genai = c(
    "synthesizers", "hallucinations", "prompting", "diffusion", "transformers",
    "embeddings", "tokenization", "latents", "checkpoints", "fine-tuning",
    "inpainting", "outpainting", "upscaling", "interpolation", "stylization",
    "remixing", "paraphrasing", "summarization", "translation", "captioning",
    "intelligence"
  ),
  ml = c(
    "regularization", "optimization", "backpropagation", "quantization",
    "distillation",
    "pruning", "ensembling", "bootstrapping", "clustering", "classification",
    "regression",
    "forecasting", "segmentation", "detection", "tracking", "reasoning",
    "grounding", "chaining", "retrieval", "augmentation"
  ),
  robotics = c(
    "actuators", "sensors", "kinematics", "dynamics", "odometry",
    "calibration", "localization", "mapping", "planning", "grasping",
    "manipulation", "locomotion", "navigation", "teleoperation",
    "haptics", "exoskeletons", "drones", "cobots", "grippers", "encoders"
  ),
  laws = c(
    "statutes", "ordinances", "prohibitions", "injunctions", "sanctions",
    "penalties", "liabilities", "compliance", "enforcement", "adjudication",
    "litigation", "arbitration", "legislation", "jurisprudence", "oversight",
    "surveillance", "censorship", "restrictions", "mandates"
  ),
  govt = c(
    "policy", "procurement", "subsidies", "grants", "initiatives", "frameworks",
    "strategies", "roadmaps", "partnerships", "sovereignty", "diplomacy",
    "infrastructure", "innovation", "competitiveness", "autonomy",
    "harmonization", "coordination", "investment", "regulation", "governance"
  ),
  global = c(
    "treaties", "conventions", "protocols", "standards", "certifications",
    "accords", "alliances", "coalitions", "forums", "summits", "councils",
    "committees", "tribunals", "arbitrators", "mediators", "negotiators",
    "diplomats", "envoys", "delegates", "stakeholders",
    "accountability", "cooperation", "inclusion"
  ),
  bias = c(
    "equity", "parity", "justice", "diversity",
    "representation", "calibration", "balance", "neutrality",
    "impartiality", "objectivity", "transparency", "accountability",
    "explainability", "interpretability", "robustness", "security",
    "privacy", "dignity", "rights", "fairness"
  )

```

```

),
misinfo = c(
  "forgeries", "fabrications", "counterfeits", "hoaxes",
  "propaganda", "disinformation", "malinformation", "satire",
  "parody", "impersonation", "spoofing", "phishing",
  "scamming", "trolling", "botting", "astroturfing", "clickbait",
  "sensationalism", "conspiracy", "pseudoscience", "influence", "validation"
),
job = c(
  "unemployment", "underemployment", "redundancy", "layoffs", "restructuring",
  "outsourcing", "offshoring", "automation", "mechanization", "computerization",
  "robotization", "reskilling", "upskilling", "retraining", "transitioning",
  "relocating", "migrating", "adapting", "pivoting", "freelancing"
),
business = c(
  "analytics", "forecasting", "optimization", "personalization",
  "recommendation", "targeting", "segmentation", "attribution", "conversion",
  "retention", "acquisition", "engagement", "monetization",
  "scalability", "efficiency", "profitability", "growth", "innovation",
  "transformation"
),
industry = c(
  "deployment", "integration", "implementation", "migration", "modernization",
  "digitization", "virtualization", "containerization", "orchestration",
  "automation", "standardization", "industrialization", "commercialization",
  "commoditization", "democratization", "decentralization", "distribution",
  "dissemination", "proliferation", "saturation"
),
productivity = c(
  "copilots", "assistants", "agents", "chatbots", "scribes", "productivity",
  "notetakers", "schedulers", "planners", "organizers", "summarizers",
  "translators", "transcribers", "researchers", "analysts",
  "designers", "coders", "writers", "editors", "presenters", "collaborators"
)
)

topics <- list(
  innovation = c("genai", "ml", "robotics"),
  regulation = c("laws", "govt", "global"),
  ethics      = c("bias", "misinfo", "job"),
  business    = c("business", "industry", "productivity")
)

topic_labels <- list(
  innovation = c("Generative AI", "Machine Learning", "Robotics"),
  regulation = c("Laws & Regulations", "Government Strategies", "Global Governance"),
  ethics      = c("Bias & Fairness", "Misinformation", "Job Displacement"),
  business    = c("AI in Business", "Industry Adoption", "AI Productivity")
)

topic_titles <- c("Innovation", "Regulation", "Ethics & Risks", "Business")
topic_colors <- c("#FF9999", "#99CCFF", "#99FF99", "#FFCC99")
bar_colors   <- c("#FF9999", "#99CCFF", "#99FF99")

```

```

file_list <- list.files(path = original_dir, pattern = "\\\\.txt$", full.names = FALSE)
if (length(file_list) == 0) stop("No .txt files found in: ", original_dir)
cat("Found", length(file_list), "text file(s):", paste(file_list, collapse = ", "),
    "\n")

count_keyword <- function(text, word) {
  str_count(text, paste0("\\b", word, "\\b"))
}

results <- data.frame(Document = file_list, stringsAsFactors = FALSE)
all_text <- ""

for (i in seq_along(file_list)) {
  raw <- paste(readLines(file.path(original_dir, file_list[i]), warn = FALSE),
    collapse = " ")
  raw <- tolower(raw)
  raw <- removeWords(raw, stopwords("en"))
  raw <- removePunctuation(raw)
  wrds <- unlist(str_split(raw, "\\s+"))
  wrds <- lemmatize_words(wrds)
  wrds <- wrds[nchar(wrds) > 0]
  proc <- paste(wrds, collapse = " ")

  all_text <- paste(all_text, proc)

  for (cat_name in names(keywords)) {
    results[i, cat_name] <- sum(sapply(keywords[[cat_name]], count_keyword, text =
    proc))
  }
}

cat("Text processing complete.\n")

topic_totals <- list()
for (t in names(topics)) {
  topic_totals[[t]] <- data.frame(
    Category = topic_labels[[t]],
    Count = as.numeric(colSums(results[, topics[[t]], drop = FALSE)))
}

overall <- data.frame(
  Topic = topic_titles,
  Count = c(
    sum(topic_totals$innovation$Count),
    sum(topic_totals$regulation$Count),
    sum(topic_totals$ethics$Count),
    sum(topic_totals$business$Count)
  )
)

```

```

overall$Percentage <- round(100 * overall$Count / sum(overall$Count), 1)
overall$Label      <- paste0(overall$Topic, "\n", overall$Percentage, "%")

cat("\n==== RESULTS BY DOCUMENT =====\n")
print(results)

for (t in names(topics)) {
  cat("\n--- TOPIC:", toupper(t), "---\n")
  print(topic_totals[[t]])
  for (i in seq_along(topics[[t]])) {
    cat_name <- topics[[t]][i]
    wc <- data.frame(
      Keyword = keywords[[cat_name]],
      Count   = sapply(keywords[[cat_name]], count_keyword, text = all_text)
    )
    wc <- wc[wc$Count > 0, ]
    wc <- wc[order(-wc$Count), ]
    cat("  Top 3 in", topic_labels[[t]][i], ":\n")
    for (k in seq_len(min(3, nrow(wc)))) {
      cat(sprintf("    %d. %s (%d)\n", k, wc$Keyword[k], wc$Count[k]))
    }
  }
}

cat("\nGenerating bar charts...\n")

for (t in names(topics)) {
  p <- ggplot(topic_totals[[t]], aes(x = Category, y = Count, fill = Category)) +
    geom_col(width = 0.7) +
    geom_text(aes(label = Count), vjust = -0.5, size = 5, fontface = "bold") +
    labs(
      title = paste("Topic:", switch(t,
        innovation = "AI Innovation & Technology",
        regulation = "AI Regulation & Government Policy",
        ethics      = "AI Ethics & Risks",
        business    = "AI in Business & Industry"
      )),
      x = "", y = "Total Keyword Mentions"
    ) +
    scale_fill_manual(values = bar_colors) +
    theme_minimal(base_size = 14) +
    theme(
      legend.position = "none",
      axis.text.x     = element_text(size = 11, face = "bold", color = "black"),
      axis.text.y     = element_text(size = 11, color = "black"),
      axis.title.y    = element_text(size = 13, face = "bold"),
      plot.title      = element_text(size = 15, face = "bold", hjust = 0.5),
      panel.grid.major.x = element_blank(),
      panel.grid.minor  = element_blank()
    )
  save_png(p, paste0("topic_", t, ".png"))
}

```

```

p_comp <- ggplot(overall, aes(x = Topic, y = Count, fill = Topic)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = Count), vjust = -0.5, size = 5, fontface = "bold") +
  labs(title = "All Topics Comparison", x = "", y = "Total Keyword Mentions") +
  scale_fill_manual(values = topic_colors) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(size = 11, face = "bold", color = "black"),
    axis.text.y = element_text(size = 11, color = "black"),
    axis.title.y = element_text(size = 13, face = "bold"),
    plot.title = element_text(size = 15, face = "bold", hjust = 0.5),
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank()
  )
save_png(p_comp, "topic_comparison.png")

p_pie <- ggplot(overall, aes(x = "", y = Count, fill = Topic)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  geom_text(
    aes(label = Label),
    position = position_stack(vjust = 0.5),
    size = 4.5,
    fontface = "bold"
  ) +
  labs(title = "Topic Coverage Distribution", x = NULL, y = NULL) +
  scale_fill_manual(values = topic_colors) +
  theme_void() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 15),
    legend.position = "none"
  )
save_png(p_pie, "topic_coverage_pie.png", width_in = 7, height_in = 7)

cat("All bar charts saved.\n")

cat("\nGenerating word clouds...\n")

all_keywords <- data.frame()
for (cat_name in names(keywords)) {
  wc <- data.frame(
    Keyword = keywords[[cat_name]],
    Count = sapply(keywords[[cat_name]], count_keyword, text = all_text)
  )
  all_keywords <- rbind(all_keywords, wc[wc$Count > 0, ])
}
all_keywords <- aggregate(Count ~ Keyword, all_keywords, sum)
all_keywords <- all_keywords[order(-all_keywords$Count), ]

for (t in names(topics)) {

```

```

wc_data <- data.frame()
for (cat_name in topics[[t]]) {
  wc <- data.frame(
    Keyword = keywords[[cat_name]],
    Count   = sapply(keywords[[cat_name]], count_keyword, text = all_text)
  )
  wc_data <- rbind(wc_data, wc[wc$Count > 0, ])
}
if (nrow(wc_data) > 0) {
  wc_data <- aggregate(Count ~ Keyword, wc_data, sum)
  wc_data <- wc_data[order(-wc_data$Count), ]
  save_wordcloud_png(wc_data,
    paste0("wordcloud_", t, ".png"),
    paste("Topic:", t))
}
}

save_wordcloud_png(all_keywords, "wordcloud_all.png", "All Topics Combined", max_words
= 60)
cat("All word clouds saved.\n")

cat("\nCreating Word document...\n")

top_keyword <- all_keywords[1, ]

add_img_safe <- function(doc, filepath, w = 6, h = 4) {
  if (file.exists(filepath)) {
    doc <- body_add_img(doc, src = filepath, width = w, height = h)
  } else {
    doc <- body_add_par(doc, paste("[Image not found:", basename(filepath), "]"),
style = "Normal")
    cat(" WARNING: Missing image:", filepath, "\n")
  }
  return(doc)
}

doc <- read_docx()

doc <- doc %>%
  body_add_par(
    "Media Trends in Artificial Intelligence Development and Regulation: A Text Mining
Study",
    style = "heading 1"
  ) %>%
  body_add_par(paste("Analysis Date:", Sys.Date()), style = "Normal") %>%
  body_add_par(paste("Files Analyzed:", length(file_list)), style = "Normal") %>%
  body_add_par(" ", style = "Normal") %>%
  body_add_par("EXECUTIVE SUMMARY", style = "heading 2") %>%
  body_add_par(paste("Total keyword mentions across all topics:", sum(overall$Count)),
style = "Normal")

for (t in names(topics)) {

```



```

w <- topic_totals[[t]]$Category[which.max(topic_totals[[t]]$Count)]
wn <- max(topic_totals[[t]]$Count)
doc <- body_add_par(doc,
                    paste(" ", toupper(t), "- Most Discussed:", w, "(", wn,
"mentions)"),
                    style = "Normal"
)
}

doc <- doc %>%
  body_add_par(" ", style = "Normal") %>%
  body_add_par("Overall Topic Ranking:", style = "Normal")

for (i in seq_len(nrow(overall[order(-overall$Count), ]))) {
  row <- overall[order(-overall$Count), ][i, ]
  doc <- body_add_par(doc,
                      paste(" ", i, ".", row$Topic, ":", row$Count, "mentions (",
row$Percentage, "%)"),
                      style = "Normal"
)
}

doc <- doc %>%
  body_add_par(" ", style = "Normal") %>%
  body_add_par(
    paste("Most Frequent Keyword Overall:", top_keyword$Keyword, "(",
top_keyword$Count, "occurrences"),
    style = "Normal"
  ) %>%
  body_add_par(" ", style = "Normal")

topic_section_titles <- c(
  innovation = "AI Innovation & Technology",
  regulation = "AI Regulation & Government Policy",
  ethics      = "AI Ethics & Risks",
  business    = "AI in Business & Industry"
)

for (idx in seq_along(names(topics))) {
  t          <- names(topics)[idx]
  t_title    <- topic_section_titles[[t]]
  t_data     <- topic_totals[[t]]
  t_cats     <- topics[[t]]
  t_names    <- topic_labels[[t]]

  doc <- doc %>%
    body_add_par(paste("TOPIC", idx, ":", t_title), style = "heading 2") %>%
    body_add_par(" ", style = "Normal") %>%
    body_add_par("Categories discussed:", style = "Normal")

  for (i in seq_len(nrow(t_data))) {

```

```

doc <- body_add_par(doc,
                    paste("  -", t_data$Category[i], ":", t_data$Count[i],
"mentions"),
                    style = "Normal"
)
}

best_cat <- t_data$Category[which.max(t_data$Count)]
best_count <- max(t_data$Count)
doc <- doc %>%
  body_add_par(
    paste("  Most discussed category:", best_cat, "with", best_count, "mentions"),
    style = "Normal"
  ) %>%
  body_add_par(" ", style = "Normal")

for (i in seq_along(t_cats)) {
  wc <- data.frame(
    Keyword = keywords[[t_cats[i]],
    Count = sapply(keywords[[t_cats[i]], count_keyword, text = all_text)
  )
  wc <- wc[wc$Count > 0, ]
  wc <- wc[order(-wc$Count), ]
  doc <- body_add_par(doc, paste("Top 3 keywords in", t_names[i], ":"), style =
"Normal")
  for (k in seq_len(min(3, nrow(wc)))) {
    doc <- body_add_par(doc,
                        paste("    ", k, ".", wc$Keyword[k], "(", wc$Count[k], ")"),
                        style = "Normal"
    )
  }
  doc <- body_add_par(doc, " ", style = "Normal")
}

img_path <- file.path(output_folder, paste0("topic_", t, ".png"))
doc <- body_add_par(doc, paste(t_title, "- Keyword Chart:"), style = "Normal")
doc <- add_img_safe(doc, img_path)
doc <- doc %>%
  body_add_par(paste("Figure", idx, ": Keyword distribution across", t_title), style
= "Normal") %>%
  body_add_par(" ", style = "Normal")
}

doc <- doc %>%
  body_add_par("OVERALL TOPIC COMPARISON", style = "heading 2") %>%
  body_add_par(" ", style = "Normal")

for (i in seq_len(nrow(overall))) {
  doc <- body_add_par(doc,
                      paste("  -", overall$Topic[i], ":", overall$Count[i], "mentions
(", overall$Percentage[i], "%)"),
                      style = "Normal"

```

```

    )
}

doc <- doc %>%
  body_add_par(" ", style = "Normal") %>%
  body_add_par("Comparison Bar Chart:", style = "Normal")
doc <- add_img_safe(doc, file.path(output_folder, "topic_comparison.png"))
doc <- doc %>%
  body_add_par("Figure 5: Total keyword mentions across all topics", style =
"Normal") %>%
  body_add_par(" ", style = "Normal") %>%
  body_add_par("TOPIC COVERAGE DISTRIBUTION", style = "heading 2") %>%
  body_add_par(" ", style = "Normal") %>%
  body_add_par("Pie chart - percentage distribution of all topics:", style = "Normal")
doc <- add_img_safe(doc, file.path(output_folder, "topic_coverage_pie.png"), w = 5, h
= 5)
doc <- doc %>%
  body_add_par("Figure 6: Topic coverage distribution", style = "Normal") %>%
  body_add_par(" ", style = "Normal") %>%
  body_add_par("INTERPRETATION", style = "heading 2") %>%
  body_add_par(" ", style = "Normal") %>%
  body_add_par(
    paste("Dominant topic:", overall$Topic[which.max(overall$Count)],
          "with", max(overall$Percentage), "% of all keyword mentions."),
    style = "Normal"
  ) %>%
  body_add_par(" ", style = "Normal") %>%
  body_add_par("Topic-Specific Insights:", style = "Normal")

for (t in names(topics)) {
  w <- topic_totals[[t]]$Category[which.max(topic_totals[[t]]$Count)]
  wn <- max(topic_totals[[t]]$Count)
  doc <- body_add_par(doc,
    paste(" -", topic_section_titles[[t]], ":", w, "with", wn,
"mentions."),
    style = "Normal"
  )
}

doc <- doc %>%
  body_add_par(" ", style = "Normal") %>%
  body_add_par(
    paste("Most frequent keyword overall: '", top_keyword$Keyword,
          "' with", top_keyword$Count, "occurrences."),
    style = "Normal"
  )

docx_path <- file.path(output_folder, "AI_Analysis_Report.docx")
print(doc, target = docx_path)
cat("Word document saved:", docx_path, "\n")

cat("\n=====\\n")

```

```

cat("                                FINAL SUMMARY                                \n")
cat("===== \n")
cat("TOPIC WINNERS: \n")
for (t in names(topics)) {
  w <- topic_totals[[t]]$Category[which.max(topic_totals[[t]]$Count)]
  wn <- max(topic_totals[[t]]$Count)
  cat(sprintf(" %-12s : %s (%d mentions)\n", t, w, wn))
}

cat("\nOVERALL TOPIC COVERAGE: \n")
for (i in seq_len(nrow(overall[order(-overall$Count), ]))) {
  row <- overall[order(-overall$Count), ][i, ]
  cat(sprintf("  %d. %-15s : %d mentions (%.1f%%)\n",
              i, row$Topic, row$Count, row$Percentage))
}

cat(sprintf("\nMOST FREQUENT KEYWORD: %s (%d occurrences)\n",
            all_keywords$Keyword[1], all_keywords$Count[1]))

cat("\nAll outputs saved to:", output_folder, "\n")
cat("\nFiles saved: \n")
for (f in c(
  "topic_innovation.png", "topic_regulation.png",
  "topic_ethics.png",     "topic_business.png",
  "topic_comparison.png", "topic_coverage_pie.png",
  "wordcloud_innovation.png", "wordcloud_regulation.png",
  "wordcloud_ethics.png",    "wordcloud_business.png",
  "wordcloud_all.png",      "AI_Analysis_Report.docx"
)) {
  status <- if (file.exists(file.path(output_folder, f))) "OK" else "MISSING"
  cat(sprintf(" [%s] %s\n", status, f))
}

```

The complete source code file (main1.R) is submitted, it contains the full keyword dictionary, all pre-processing steps, the classification and counting logic, all visualization code, and the Word document generation code.

APPENDIX B: KEYWORD DICTIONARY – FULL LISTING

Table B1 below provides the full listing of all keywords used in the analysis, organized by topic and subcategory.

Topic	Subcategory	Keywords
Innovation	Generative AI	synthesizers, hallucinations, prompting, diffusion, transformers, embeddings, tokenization, latents, checkpoints, fine-tuning, inpainting, outpainting, upscaling, interpolation, stylization, remixing, paraphrasing, summarization, translation, captioning, intelligence
Innovation	Machine Learning	regularization, optimization, backpropagation, quantization, distillation, pruning, ensembling, bootstrapping, clustering, classification, regression, forecasting, segmentation, detection, tracking, reasoning, grounding, chaining, retrieval, augmentation
Innovation	Robotics	actuators, sensors, kinematics, dynamics, odometry, calibration, localization, mapping, planning, grasping, manipulation, locomotion, navigation, teleoperation, haptics, exoskeletons, drones, cobots, grippers, encoders
Regulation	Laws & Regulations	statutes, ordinances, prohibitions, injunctions, sanctions, penalties, liabilities, compliance, enforcement, adjudication, litigation, arbitration, legislation, jurisprudence, oversight, surveillance, censorship, restrictions, mandates
Regulation	Government Strategies	policy, procurement, subsidies, grants, initiatives, frameworks, strategies, roadmaps, partnerships, sovereignty, diplomacy, infrastructure, innovation,

Topic	Subcategory	Keywords
Regulation	Global Governance	competitiveness, autonomy, harmonization, coordination, investment, regulation, governance treaties, conventions, protocols, standards, certifications, accords, alliances, coalitions, forums, summits, councils, committees, tribunals, arbitrators, mediators, negotiators, diplomats, envoys, delegates, stakeholders, accountability, cooperation, inclusion
Ethics	Bias & Fairness	equity, parity, justice, diversity, representation, calibration, balance, neutrality, impartiality, objectivity, transparency, accountability, explainability, interpretability, robustness, security, privacy, dignity, rights, fairness
Ethics	Misinformation	forgeries, fabrications, counterfeits, hoaxes, propaganda, disinformation, malinformation, satire, parody, impersonation, spoofing, phishing, scamming, trolling, botting, astroturfing, clickbait, sensationalism, conspiracy, pseudoscience, influence, validation
Ethics	Job Displacement	unemployment, underemployment, redundancy, layoffs, restructuring, outsourcing, offshoring, automation, mechanization, computerization, robotization, reskilling, upskilling, retraining, transitioning, relocating, migrating, adapting, pivoting, freelancing
Business	AI in Business	analytics, forecasting, optimization, personalization, recommendation, targeting, segmentation, attribution, conversion, retention, acquisition, engagement, monetization, scalability,

Topic	Subcategory	Keywords
		efficiency, profitability, growth, innovation, transformation
Business	Industry Adoption	deployment, integration, implementation, migration, modernization, digitization, virtualization, containerization, orchestration, automation, standardization, industrialization, commercialization, commoditization, democratization, decentralization, distribution, dissemination, proliferation, saturation
Business	AI Productivity	copilots, assistants, agents, chatbots, scribes, productivity, notetakers, schedulers, planners, organizers, summarizers, translators, transcribers, researchers, analysts, designers, coders, writers, editors, presenters, collaborators

Table B1: Full keyword dictionary organised by topic and subcategory